IMAT 모의고사 2회

60 questions · 100 minutes · +1.5 / −0.4 / 0

Section 1 — Reading Skills & General Knowledge (Q1–4)

Passage for Q1–2

> Antibiotic resistance is often described as a problem of overuse in human medicine. Yet in many countries more antibiotics are given to farm animals than to people, frequently to promote growth rather than to treat disease. Resistant bacteria and resistance genes can move from animals to humans through food, water and direct contact. Some countries that restricted growth-promoting antibiotics saw resistance in animal bacteria fall within a few years, although effects on human infections were harder to measure.

1. What is the main point of the passage?

A. Resistance genes cannot transfer between species.
B. Restricting antibiotics in farming has eliminated resistance in humans.
C. Agricultural use of antibiotics is a significant, and sometimes overlooked, contributor to resistance.
D. Human overuse of antibiotics is not a problem.
E. Antibiotic resistance is caused only by agriculture.

2. According to the passage, what happened after some countries restricted growth-promoting antibiotics?

A. Antibiotic use in humans increased.
B. Resistance genes stopped moving through water.
C. Human infections immediately decreased.
D. Farm productivity collapsed.
E. Resistance in bacteria from animals declined.

3. Which of the following Italian cities was the capital of the Kingdom of Italy immediately before Rome became capital in 1871?

A. Venice
B. Naples
C. Turin
D. Milan
E. Florence

4. The discovery of the structure of DNA in 1953 is associated with Watson and Crick, but which scientist's X-ray diffraction images were crucial to it?

A. Marie Curie
B. Rosalind Franklin
C. Lise Meitner
D. Dorothy Hodgkin
E. Barbara McClintock

Section 2 — Logical Reasoning & Problem Solving (Q5–9)

5. "If a patient has a fever, the doctor orders a blood test. The doctor did not order a blood test." Which conclusion follows?

A. The doctor made a mistake.
B. The patient has an infection.
C. The patient has a fever.
D. Nothing can be concluded.
E. The patient does not have a fever.

6. A train leaves at 08:45 and travels 270 km at an average speed of 120 km/h. At what time does it arrive?

A. 10:55
B. 11:30
C. 11:15
D. 10:45
E. 11:00

7. In a class of 30 students, 18 study biology, 14 study chemistry and 5 study neither. How many study both?

A. 9
B. 7
C. 11
D. 3
E. 5

8. Which number completes the pattern: 2, 6, 12, 20, 30, __?

A. 48
B. 36
C. 42
D. 40
E. 44

9. A medicine must be taken every 8 hours. The first dose is taken at 22:00 on Monday. When is the 5th dose taken?

A. 22:00 Tuesday
B. 06:00 Tuesday
C. 22:00 Wednesday
D. 06:00 Wednesday
E. 14:00 Wednesday

Section 3 — Biology (Q10–32)

10. Which of the following is found in plant cells but not in animal cells?

A. Chloroplasts
B. Ribosomes
C. Golgi apparatus
D. Mitochondria
E. Plasma membrane

11. The enzyme that joins Okazaki fragments during DNA replication is:
A. DNA ligase
B. DNA polymerase III
C. Primase
D. Topoisomerase
E. Helicase

12. Which statement about the cell cycle is correct?
A. Mitosis takes up most of the cell cycle.
B. DNA is replicated during G1.
C. Cytokinesis occurs during prophase.
D. Cells in G0 are actively dividing.
E. DNA is replicated during S phase.

13. How many net ATP molecules are produced directly by glycolysis per molecule of glucose?
A. 2
B. 1
C. 32
D. 4
E. 38

14. A man with blood group AB and a woman with blood group O have a child. Which statement is correct?
A. The child may have blood group O.
B. The child must have blood group AB.
C. The child may have blood group AB.
D. The child can only have blood group A or B.
E. The child must have blood group A.

15. Where does the Calvin cycle take place?
A. Stroma of the chloroplast
B. Mitochondrial matrix
C. Thylakoid membrane
D. Nucleus
E. Cytoplasm

16. Which structure prevents food from entering the trachea during swallowing?
A. Pharynx
B. Soft palate
C. Larynx
D. Uvula
E. Epiglottis

17. Which of these molecules is a steroid hormone?
A. Oxytocin
B. Cortisol
C. Glucagon
D. Insulin
E. Adrenaline

18. Which ion entering the presynaptic terminal triggers neurotransmitter release?

A. Na^+
B. K^+
C. Mg^{2+}
D. Cl^-
E. Ca^{2+}

19. In which chamber of the heart does blood have the highest pressure during systole?

A. Right ventricle
B. Left ventricle
C. Left atrium
D. All chambers have equal pressure
E. Right atrium

20. A test cross of a plant with dominant phenotype gives 50% dominant and 50% recessive offspring. What is the genotype of the tested plant?

A. aa
B. Cannot be determined
C. AA or Aa equally likely
D. Aa
E. AA

21. Which hormone stimulates the conversion of glycogen to glucose in the liver?

A. Calcitonin
B. Aldosterone
C. Glucagon
D. Melatonin
E. Insulin

22. Which base is found in RNA but not in DNA?

A. Guanine
B. Adenine
C. Uracil
D. Cytosine
E. Thymine

23. Vaccination mainly works by:

A. providing passive immunity through antibodies from another organism.
B. preventing pathogens from entering the body.
C. stimulating the production of memory cells.
D. directly killing pathogens.
E. increasing the number of red blood cells.

24. The hormone aldosterone primarily acts on the kidney to:

A. decrease Na^+ reabsorption.
B. increase water excretion.
C. increase Na^+ reabsorption and K^+ secretion.
D. increase urea synthesis.
E. increase glucose reabsorption.

25. A human cell in metaphase of mitosis contains how many chromatids?
A. 69
B. 46
C. 92
D. 184
E. 23

26. Which level of protein structure is defined by the sequence of amino acids?
A. Quaternary
B. Domain
C. Secondary
D. Primary
E. Tertiary

27. The deletion of a single nucleotide in a coding region usually causes:
A. a change in the promoter.
B. removal of one amino acid only.
C. no change in the protein.
D. a frameshift that alters all downstream codons.
E. a single amino acid substitution.

28. Which transport process requires ATP directly?
A. Primary active transport
B. Simple diffusion
C. Diffusion through channels
D. Facilitated diffusion
E. Osmosis

29. Viruses are not considered cells because they:
A. have proteins.
B. contain nucleic acids.
C. can mutate.
D. cannot reproduce outside a host cell and lack their own metabolism.
E. infect bacteria.

30. Bile helps digestion by:
A. absorbing vitamin B12.
B. digesting starch.
C. emulsifying fats.
D. hydrolysing proteins.
E. neutralising stomach acid only.

31. Which codon serves as the start codon in most organisms?
A. AUG
B. UAA
C. GUG
D. UAG
E. UGA

32. Which of the following is an example of natural selection?

A. Giraffes stretching their necks to reach leaves and passing longer necks to offspring.
B. A person becoming stronger by training.
C. Producing hybrid corn.
D. Bacteria surviving antibiotic treatment because of a resistance gene, then multiplying.
E. Breeding dogs for herding.

Section 4 — Chemistry (Q33–47)

33. What is the molar mass of H_2SO_4? (H = 1, S = 32, O = 16)

A. 49 g/mol
B. 96 g/mol
C. 100 g/mol
D. 64 g/mol
E. 98 g/mol

34. What is the oxidation number of chromium in $Cr_2O_7^{2-}$?

A. +7
B. +6
C. +12
D. +3
E. +4

35. What is the pH of a 0.01 mol/L NaOH solution at 25 °C?

A. 10
B. 2
C. 7
D. 14
E. 12

36. Down group 1, from Li to Cs, which property increases?

A. Number of valence electrons
B. Melting point
C. Electronegativity
D. First ionisation energy
E. Atomic radius

37. Which molecule is non-polar despite having polar bonds?

A. SO_2
B. NH_3
C. CO_2
D. H_2O
E. HCl

38. How many grams of water are produced when 4 g of H_2 reacts completely with excess O_2? (H = 1, O = 16)

A. 18 g
B. 9 g
C. 36 g
D. 32 g
E. 72 g

39. Oxidation of a primary alcohol under strong conditions gives:

A. a tertiary alcohol.
B. a carboxylic acid.
C. an ether.
D. an alkene.
E. a ketone.

40. What volume of 2 mol/L HCl is needed to prepare 500 mL of 0.4 mol/L HCl?

A. 50 mL
B. 100 mL
C. 200 mL
D. 400 mL
E. 250 mL

41. Which bond type forms when electrons are transferred from one atom to another?

A. Polar covalent
B. Hydrogen
C. Non-polar covalent
D. Ionic
E. Metallic

42. What is the conjugate base of $H_2PO_4^-$?

A. H_3PO_4
B. PO_4^{3-}
C. OH^-
D. HPO_4^{2-}
E. H_2O

43. Increasing temperature increases the rate of a reaction mainly because:

A. more molecules have energy greater than the activation energy.
B. the activation energy decreases.
C. the reaction becomes exothermic.
D. the concentration of reactants increases.
E. the equilibrium constant always increases.

44. Which compound is an aldehyde?

A. $CH_3CH_2NH_2$
B. CH_3CH_2CHO
C. $CH_3CH_2OCH_3$
D. CH_3COCH_3
E. CH_3CH_2COOH

45. What is the electron configuration of a neutral oxygen atom?
A. $1s^2\ 2s^2\ 2p^4$
B. $1s^2\ 2s^2\ 2p^2$
C. $1s^2\ 2s^2\ 2p^6$
D. $1s^2\ 2p^6$
E. $1s^2\ 2s^4\ 2p^2$

46. In an electrolytic cell, oxidation occurs at the:
A. both electrodes equally.
B. anode.
C. cathode.
D. salt bridge.
E. electrolyte.

47. Which of the following is an amino acid functional group pair?
A. Thiol and nitrile
B. Amine and carboxyl
C. Ketone and ester
D. Ether and amide
E. Hydroxyl and aldehyde

Section 5 — Physics & Mathematics (Q48–60)

48. A ball is dropped from rest. Ignoring air resistance ($g = 10\ m/s^2$), what is its speed after 3 s?
A. 15 m/s
B. 3 m/s
C. 45 m/s
D. 30 m/s
E. 10 m/s

49. A force of 20 N moves an object 5 m in the direction of the force. How much work is done?
A. 4 J
B. 100 J
C. 15 J
D. 200 J
E. 25 J

50. Three 6 Ω resistors are connected in series to a 9 V battery. What current flows?
A. 1.5 A
B. 0.5 A
C. 4.5 A
D. 3 A
E. 1 A

51. The speed of light in vacuum is approximately 3×10^8 m/s. What is the wavelength of light with frequency 6×10^{14} Hz?

A. 5×10^{-7} m
B. 2×10^{-6} m
C. 5×10^{-6} m
D. 1.8×10^{23} m
E. 2×10^{6} m

52. An ideal gas at constant temperature has its volume halved. Its pressure:

A. quadruples.
B. stays the same.
C. becomes zero.
D. halves.
E. doubles.

53. A 1000 kg car moves at 10 m/s. What is its momentum?

A. 50 000 kg·m/s
B. 100 000 kg·m/s
C. 1000 kg·m/s
D. 100 kg·m/s
E. 10 000 kg·m/s

54. What is the power of a device that uses 1200 J in 1 minute?

A. 20 W
B. 72 000 W
C. 1200 W
D. 10 W
E. 60 W

55. If $3^x = 81$, what is x?

A. 4
B. 9
C. 81
D. 3
E. 27

56. In a right triangle, the legs are 6 cm and 8 cm. What is the hypotenuse?

A. 12 cm
B. 48 cm
C. 10 cm
D. 7 cm
E. 14 cm

57. A bag contains 3 red and 5 blue balls. Two are drawn without replacement. What is the probability that both are red?

A. 3/8
B. 9/64
C. 3/28
D. 3/14
E. 1/4

58. What is the slope of the line through (1, 2) and (4, 11)?

A. 2
B. 13/5
C. 9
D. 3
E. 1/3

59. Simplify $(x^2 - 9)/(x - 3)$ for $x \neq 3$.

A. $x + 9$
B. $3x$
C. $x - 3$
D. $x + 3$
E. $x^2 - 3$

60. An object of mass 5 kg experiences a net force of 15 N. What is its acceleration?

A. 3 m/s^2
B. 75 m/s^2
C. 10 m/s^2
D. 0.33 m/s^2
E. 20 m/s^2

정답 및 해설

번호	정답	해설
1	C	농업 사용도 내성의 주요(간과된) 원인이라는 것이 요지.
2	E	동물 세균의 내성이 몇 년 안에 감소. 인체 효과는 측정 어려움.
3	E	피렌체(1865–1871).
4	B	로절린드 프랭클린의 X선 회절 사진(Photo 51).
5	E	대우(modus tollens): 검사 안 함 → 열 없음.
6	E	270/120 = 2.25 h = 2시간 15분 → 08:45 + 2:15 = 11:00.
7	B	30−5=25 = 18+14−x → x=7.
8	C	n(n+1): 6×7=42.
9	D	22:00 월 → 06:00 화 → 14:00 화 → 22:00 화 → 06:00 수(5번째).
10	A	엽록체(그 외 세포벽, 큰 액포).
11	A	DNA 리가아제.
12	E	S기에 DNA 복제.
13	A	4개 생성 − 2개 소모 = 순 2 ATP.
14	D	AB × OO → A 또는 B만 가능, O와 AB 불가.
15	A	엽록체 스트로마.
16	E	후두개(epiglottis).
17	B	코르티솔(부신피질, 스테로이드).
18	E	Ca^{2+} 유입 → 시냅스 소포 방출.
19	B	좌심실(체순환 압력).
20	D	1:1 → 이형접합 Aa.
21	C	글루카곤 → 글리코겐 분해.
22	C	우라실.
23	C	기억세포 형성(능동면역).
24	C	알도스테론: Na^+ 재흡수↑, K^+ 분비↑.
25	C	46 염색체 × 2 염색분체 = 92.
26	D	1차 구조 = 아미노산 서열.
27	D	틀이동 돌연변이.
28	A	1차 능동수송(예: Na^+/K^+-ATPase).
29	D	숙주 밖 증식 불가, 독자적 대사 없음.
30	C	지방 유화.
31	A	AUG.
32	D	변이 존재 → 선택압 → 생존·번식. A는 라마르크설.
33	E	2+32+64 = 98.
34	B	2x + 7(−2) = −2 → x = +6.
35	E	pOH 2 → pH 12.

번호	정답	해설
36	E	원자 반지름 증가.
37	C	CO_2 직선형 → 쌍극자 상쇄.
38	C	4 g H_2 = 2 mol → H_2O 2 mol = 36 g.
39	B	1차 알코올 → 알데하이드 → 카복실산.
40	B	$C_1V_1=C_2V_2$ → V_1 = 0.4×500/2 = 100 mL.
41	D	이온결합.
42	D	H^+ 하나 제거 → HPO_4^{2-}.
43	A	활성화에너지 이상 분자 비율 증가.
44	B	프로판알(–CHO).
45	A	8개 전자: $1s^2\ 2s^2\ 2p^4$.
46	B	산화는 항상 anode.
47	B	아미노기 + 카복실기.
48	D	v = gt = 30 m/s.
49	B	W = Fd = 100 J.
50	B	18 Ω → 9/18 = 0.5 A.
51	A	$\lambda = c/f = 3\times10^8/6\times10^{14} = 5\times10^{-7}$ m.
52	E	보일 법칙.
53	E	p = mv = 10 000.
54	A	1200/60 = 20 W.
55	A	$3^4 = 81$.
56	C	6-8-10.
57	C	3/8 × 2/7 = 6/56 = 3/28.
58	D	(11−2)/(4−1) = 3.
59	D	(x−3)(x+3)/(x−3).
60	A	$a = F/m = 3\ m/s^2$.